ELECTRODYNAMICS II · SEMESTER 5

# Fields in matter

Module 1 · Answer key

40 worked answers. Each answer names the prepare topic to revise first. Mark splits are suggested study rubrics.

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## Section A

### Q1 Model · syllabus-based

**How can an electrically neutral dielectric respond to a static electric field?**

KNOW FIRST

T1 Dielectrics and induced dipoles P1 School physics bridge M1 Mathematics from the beginning

Its positive and negative charge centres shift slightly relative to one another, or existing molecular dipoles turn. Charges remain bound rather than crossing the whole material.

**ANSWER** **Neutrality does not prevent polarization.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

### Q2 Legacy PYQ N25 Q5 · expanded

**Define atomic polarizability and give its SI unit.**

KNOW FIRST

T1 Dielectrics and induced dipoles P1 School physics bridge M1 Mathematics from the beginning

For a weak local field, $\mathbf{p} = \alpha \mathbf{E}_{\text{local}}$. Thus $[\alpha] = (\mathrm{C\,m})/(\mathrm{V/m}) = \mathrm{C\,m^2/V}$. The atom is isotropic in this scalar linear model; α describes an individual atom rather than the whole sample.

**ANSWER** **Polarizability is induced dipole moment per unit local field.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q3 Model · syllabus-based

**A permanent dipole has moment $2 \times 10^{-29}$ C m in a field $3 \times 10^{5}$ V/m at 30°. Find the torque magnitude.**

KNOW FIRST

T2 Permanent dipoles: torque and force P1 School physics bridge M1 Mathematics from the beginning

$\tau = pE \sin\theta = (2 \times 10^{-29})(3 \times 10^{5})(0.5) = 3 \times 10^{-24}$ N m. The torque turns p toward E.

**ANSWER** $\tau = 3 \times 10^{-24}$ **N m.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q4 Model · syllabus-based

**Which orientation of a permanent dipole in a uniform field is stable? Use energy.**

KNOW FIRST

T2 Permanent dipoles: torque and force P1 School physics bridge M1 Mathematics from the beginning

$U = -pE\cos\theta$ is minimum at $\theta = 0$. A small angular displacement produces a restoring torque. The antiparallel orientation is an energy maximum.

**ANSWER** **Parallel alignment is stable.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q5 Model · syllabus-based

**Define polarization and distinguish it from dipole moment.**

KNOW FIRST

T3 Polarization as a density P1 School physics bridge M1 Mathematics from the beginning

P is electric dipole moment per unit volume (C/m²). The moment p of one dipole has units C m. P is a bulk average and can vanish through random orientation.

**ANSWER** $\mathbf{P} = \Delta\mathbf{p}/\Delta\tau$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q6 Model · syllabus-based

**A material has $5 \times 10^{27}$ aligned dipoles per m³, each of moment $2 \times 10^{-30}$ C m. Find P.**

KNOW FIRST

T3 Polarization as a density P1 School physics bridge M1 Mathematics from the beginning

Multiply moment per dipole by dipoles per volume: $P = np = 10^{-2}$ C/m², directed along the common dipole direction.

**ANSWER** $P = 0.010$ **C/m².**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q7 Model · syllabus-based

**A face normal makes 60° with uniform $\mathbf{P}$, whose magnitude is 0.04 C/m². Find the bound surface charge density.**

KNOW FIRST

T4 Bound charge and the field of polarization P1 School physics bridge M1 Mathematics from the beginning

$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = P\cos 60^\circ = 0.020$ C/m². This sign assumes the stated outward normal.

**ANSWER** $\sigma_b = +0.020$ **C/m².**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q8 Legacy PYQ N25 Q10 · expanded

**What is the physical interpretation of bound charges? Include the case of uniform polarization.**

KNOW FIRST

T4 Bound charge and the field of polarization P1 School physics bridge M1 Mathematics from the beginning

Polarization separates positive and negative charge within atoms or molecules, leaving unmatched dipole ends at boundaries and wherever the dipole density varies. These

charges remain associated with the material, unlike mobile free charge. For uniform P, $\rho_b = 0$ inside but $\sigma_b = P \cdot \hat{n}$ may remain at its surface.

**ANSWER** **Bound charges are the macroscopic result of displaced or aligned microscopic dipoles.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q9 Model · syllabus-based

**Define electric displacement and state its SI unit.**

KNOW FIRST

T5 Displacement and free charge P1 School physics bridge M1 Mathematics from the beginning

$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$. Both terms have units C/m². Its closed-surface flux equals enclosed free charge.

**ANSWER** **D is measured in C/m².**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q10 Model · syllabus-based

**Which charge appears in Gauss's law for D, and which appears in Gauss's law for E?**

KNOW FIRST

T5 Displacement and free charge P1 School physics bridge M1 Mathematics from the beginning

D flux counts $Q_f$. The electric field law is $\epsilon_0 \oint \mathbf{E} \cdot d\mathbf{a} = Q_f + Q_b$. Bound charge is still physically present.

**ANSWER** **D counts free charge; E counts total charge.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q11 Model · syllabus-based

**State the boundary condition for the normal component of D, with your normal convention.**

KNOW FIRST

T6 Electric boundary conditions P1 School physics bridge M1 Mathematics from the beginning

Let n point from medium 1 to 2. Then $(\mathbf{D}_2 - \mathbf{D}_1) \cdot \hat{\mathbf{n}} = \sigma_f$. This follows from a thin Gaussian pillbox.

**ANSWER** $D_{2n} - D_{1n} = \sigma_f$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q12 Model · syllabus-based

**Why is tangential E continuous across an electrostatic interface?**

KNOW FIRST

T6 Electric boundary conditions P1 School physics bridge M1 Mathematics from the beginning

The electrostatic circulation $\oint \mathbf{E} \cdot d\mathbf{l}$ is zero. For a shrinking rectangular loop crossing the surface only its tangential sides survive, so $E_{2t} = E_{1t}$.

**ANSWER** **Tangential E has no electrostatic jump.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q13 Model · syllabus-based

**A linear dielectric has $\epsilon_r = 5$. Find its electric susceptibility.**

KNOW FIRST

T7 Linear dielectrics and capacitors P1 School physics bridge M1 Mathematics from the beginning

$\epsilon_r = 1 + \chi_e$, so $\chi_e = 4$. Both relative permittivity and susceptibility are dimensionless.

**ANSWER** $\chi_e = 4$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q14 Model · syllabus-based

**A capacitor stays connected to an ideal battery while a dielectric completely fills it. How do capacitance and free charge change?**

KNOW FIRST

T7 Linear dielectrics and capacitors P1 School physics bridge M1 Mathematics from the beginning

For unchanged geometry, C increases by $\epsilon_r$. The battery fixes V, so $Q = CV$ also increases by $\epsilon_r$. Away from edges $E = V/d$ is unchanged.

**ANSWER** **Both C and Q multiply by $\epsilon_r$.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q15 Legacy PYQ N25 Q9

**Distinguish paramagnets and diamagnets.**

KNOW FIRST

T8 Magnetic response of atoms P1 School physics bridge M1 Mathematics from the beginning

Diamagnets have a weak induced response opposing the applied field, with $\chi_m < 0$ and $\mu_r < 1$. Paramagnets have pre-existing moments that partially align with the field, giving small $\chi_m > 0$ and $\mu_r > 1$. Thermal disorder competes with paramagnetic alignment.

**ANSWER** **Diamagnets oppose weakly; paramagnets align weakly.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q16 Model · syllabus-based

**Why does a paramagnet have weak alignment rather than perfect alignment at ordinary temperature?**

KNOW FIRST

T8 Magnetic response of atoms P1 School physics bridge M1 Mathematics from the beginning

Permanent atomic moments tend to align with an applied field, but thermal agitation disorders them. The resulting average alignment is small and normally decreases when temperature rises at fixed weak field.

**ANSWER** **Field alignment competes with thermal disorder.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q17 Legacy PYQ N25 Q11 · expanded

**Define magnetic vector potential. How can it describe the field of a magnetized object?**

KNOW FIRST

T10 Bound currents and magnetized objects P1 School physics bridge M1 Mathematics from the beginning

The vector potential A is defined by $\mathbf{B} = \nabla \times \mathbf{A}$ and has units T m. In magnetostatics a localized magnetized object can be replaced by $\mathbf{J}_b = \nabla \times$

$M$ and $\mathbf{K}_b = M \times \hat{n}$ in the usual current-source potential integrals. A is not unique: adding a gradient leaves its curl unchanged.

**ANSWER** **B is curl A; equivalent bound currents may be used as A sources.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q18 Legacy PYQ N25 Q18 · adapted marks

**Discuss briefly the bound currents in a magnetized medium, emphasizing their physical interpretation.**

KNOW FIRST

T10 Bound currents and magnetized objects P1 School physics bridge M1 Mathematics from the beginning

Microscopic magnetic dipoles can be modelled as current loops. Opposite currents on shared interior edges cancel for uniform M, leaving a surface current $K_b = M \times \hat{n}$. A spatially nonuniform magnetization can also leave $J_b = \nabla \times M$ in the bulk. These are equivalent macroscopic currents, not freely drifting conduction carriers.

**ANSWER** **Uncancelled microscopic loops produce bound surface and volume currents.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q19 Model · syllabus-based

**Define H and state its magnetostatic circulation law.**

KNOW FIRST

T11 The auxiliary field H P1 School physics bridge M1 Mathematics from the beginning

$\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$. Its circulation is $\oint \mathbf{H} \cdot d\mathbf{l} = I_{f,\text{enc}}$, using the right-hand orientation of the loop and current.

**ANSWER** **Only enclosed free current appears in the H circulation law.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Q20 Legacy PYQ N25 Q3 · expanded

**Define magnetic susceptibility and relate it to relative permeability for a linear isotropic material.**

KNOW FIRST

T12 Magnetic susceptibility and permeability P1 School physics bridge M1 Mathematics from the beginning

Magnetic susceptibility is the dimensionless coefficient $\chi_m$ in $\mathbf{M} = \chi_m \mathbf{H}$. Since $B = \mu_0(H + M)$, $\mu_r = 1 + \chi_m$. For $\chi_m = 0.002$, $\mu_r = 1.002$. Ferromagnets need not obey one constant susceptibility.

**ANSWER** **Magnetic susceptibility measures magnetization per unit H in the linear regime.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

## Section B

### Q21 Model · syllabus-based

**Explain remanence, coercivity and the role of domain behaviour in an M-H hysteresis loop.**

KNOW FIRST

T13 Ferromagnetism and memory P1 School physics bridge M1 Mathematics from the beginning

1. A ferromagnet contains domains with cooperative moment alignment.
   Why: Domains need not initially point in the same direction.
2. Applied H moves walls and rotates domains.
   Why: Favourable orientations grow.
3. After H returns to zero, pinning can leave nonzero M, called remanence.
   Why: The path depends on previous magnetic history.
4. A reverse H is needed to reduce M to zero; its magnitude on that branch is the coercivity.
   Why: The two intercepts describe different properties.

5. A narrow loop suits repeated cycling, whereas substantial coercivity helps a permanent magnet retain its state.
   Why: Applications depend on loss and resistance to demagnetization.

**ANSWER** **Remanence measures retained M; coercivity measures the required reverse H.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q22 Model · syllabus-based

**Explain the effect of heating a ferromagnet through the Curie temperature, and distinguish the result from diamagnetism.**

KNOW FIRST

T13 Ferromagnetism and memory P1 School physics bridge M1 Mathematics from the beginning

1. Below the Curie temperature, interactions can maintain spontaneous alignment within domains.
   Why: Cooperative order survives thermal fluctuations.
2. Raising temperature increases disorder and reduces spontaneous magnetization.
   Why: Thermal agitation competes with ordering interactions.
3. Above the transition, spontaneous ferromagnetic order disappears and the material is paramagnetic.
   Why: It may still align weakly in an applied field.

4. This is not diamagnetism: a paramagnetic response is positive and comes from existing moments, whereas the diamagnetic contribution opposes the field.
   Why: The microscopic mechanisms differ.

**ANSWER** **Above its Curie temperature a ferromagnet loses spontaneous order and becomes paramagnetic.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q23 Model · syllabus-based

**Model an atom as a point nucleus +q in a uniform spherical cloud −q of radius a. Derive its polarizability for a small nucleus displacement d.**

KNOW FIRST

T1 Dielectrics and induced dipoles P1 School physics bridge M1 Mathematics from the beginning

1. The cloud density is $-3q/(4\pi a^3)$ and its interior field at displacement d is $-qd/(4\pi\epsilon_0 a^3)$.
   Why: Gauss's law for a uniform sphere gives a field proportional to radius.

2. At equilibrium $E = qd/(4\pi\epsilon_0 a^3)$.
   Why: The applied and restoring forces cancel.

3. Use $p = qd$ to obtain $p = 4\pi\epsilon_0 a^3 E$.
   Why: Dipole moment is charge times separation.

**ANSWER** $\alpha = 4\pi\epsilon_0 a^3$ **in this classical model.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q24 Model · syllabus-based

**A fixed dipole $\mathbf{p} = p\hat{\mathbf{x}}$ is in $\mathbf{E} = kx^2\hat{\mathbf{x}}$. Find its force at x. Is there a torque?**

KNOW FIRST

T2 Permanent dipoles: torque and force P1 School physics bridge M1 Mathematics from the beginning

1. The positive and negative ends sample different fields. To first order $F_x = p\,dE_x/dx$.
   Why: Subtract their two electric forces.
2. Since $d(kx^2)/dx = 2kx$, $\mathbf{F} = 2pkx\hat{\mathbf{x}}$.
   Why: p is fixed, so only the field is differentiated.
3. The torque is zero because p and E are collinear.
   Why: Their cross product vanishes.

**ANSWER** $\mathbf{F} = 2pkx\hat{\mathbf{x}}$; $\boldsymbol{\tau} = 0$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q25 Model · syllabus-based

**A sphere of radius R has $\mathbf{P} = a\mathbf{r}$ inside and zero outside. Find its bound charge densities and total bound charge.**

KNOW FIRST

T4 Bound charge and the field of polarization P1 School physics bridge M1 Mathematics from the beginning

1. $\nabla \cdot \mathbf{r} = 3$, hence $\rho_b = -3a$.
   Why: Each Cartesian coordinate contributes one derivative.

2. At $r = R$, $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{r}} = aR$.
   Why: The sphere's normal is radial.

3. $Q_{b,v} = (-3a)(4\pi R^3/3) = -4\pi a R^3$ and $Q_{b,s} = aR(4\pi R^2) = 4\pi a R^3$.
   Why: Integrate the constant densities over volume and area.

**ANSWER** $\rho_b = -3a$, $\sigma_b = aR$; **total bound charge is zero.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q26 Model · syllabus-based

**An infinite slab is uniformly polarized perpendicular to its faces, with no free charge. Find E inside and outside.**

KNOW FIRST

T4 Bound charge and the field of polarization P1 School physics bridge M1 Mathematics from the beginning

1. The faces carry $-P$ and $+P$.
   Why: Use outward normals in $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$.
2. Each infinite sheet produces field magnitude $P/(2\epsilon_0)$.
   Why: Apply Gauss's law to one sheet.
3. Between the sheets the fields add from positive to negative, opposing P; outside they cancel.
   Why: Superpose the two sheet fields.

**ANSWER** Inside $\mathbf{E} = -\mathbf{P}/\epsilon_0$; outside E=0.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q27 Legacy PYQ N25 Q19 · expanded

**A long straight wire with uniform free line charge λ is surrounded by rubber insulation out to radius a. Find the electric displacement, and state E if the rubber has permittivity ε and the exterior is vacuum.**

KNOW FIRST

T5 Displacement and free charge P1 School physics bridge M1 Mathematics from the beginning

1. Assume an infinitely long wire and cylindrical symmetry. A coaxial Gaussian cylinder of radius s and length ℓ has constant radial D on its curved face.
   Why: Translation and rotation symmetry remove axial and angular variation.

2. The cap flux is zero, and curved flux is $D(2\pi s\ell)$.
   Why: D is radial.

3. Enclosed free charge is $\lambda\ell$, so $D = \lambda/(2\pi s)$.
   Why: Gauss's law for D counts free charge only.

4. Thus D has the same expression in rubber and outside. In rubber, $E = D/\epsilon$; outside, $E = D/\epsilon_0$.
   Why: The constitutive relation changes at s=a even though free-charge flux does not.

**ANSWER** $\mathbf{D} = \lambda\hat{s}/(2\pi s)$ **outside the wire; E changes with the local permittivity.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q28 Model · syllabus-based

**At an uncharged interface between isotropic dielectrics, derive the relation between field angles $\theta_1, \theta_2$ measured from the normal.**

KNOW FIRST

T6 Electric boundary conditions P1 School physics bridge M1 Mathematics from the beginning

1. Tangential continuity gives $E_1 \sin\theta_1 = E_2 \sin\theta_2$.
   Why: Electrostatic circulation vanishes.
2. Normal D continuity gives $\epsilon_1 E_1 \cos\theta_1 = \epsilon_2 E_2 \cos\theta_2$.
   Why: There is no free surface charge.
3. Divide the first equation by the second.
   Why: Eliminate the field magnitudes.

**ANSWER** $\tan\theta_2 / \tan\theta_1 = \epsilon_2/\epsilon_1$ **where the ratio is defined.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q29 Model · syllabus-based

**A fully filled parallel-plate capacitor has area $0.010$ m², separation 1 mm and $\epsilon_r = 4$. At 100 V find C, free Q and P.**

KNOW FIRST

T7 Linear dielectrics and capacitors P1 School physics bridge M1 Mathematics from the beginning

1. $C = \epsilon_0 \epsilon_r A/d = 3.542 \times 10^{-10}$ F.
   Why: Ignore fringe fields.
2. $Q_f = CV = 3.542 \times 10^{-8}$ C and $E = V/d = 10^5$ V/m.
   Why: The battery sets the potential difference.
3. $P = \epsilon_0(\epsilon_r - 1)E = 2.656 \times 10^{-6}$ C/m².
   Why: Use the linear constitutive law.

**ANSWER** $C = 354.2$ pF; $Q_f = 35.42$ nC; $P = 2.656$ µC/m².

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q30 Model · syllabus-based

**A charged capacitor is disconnected from its battery before a dielectric of relative permittivity k is inserted fully. Derive the ratios of V and stored energy.**

KNOW FIRST

T7 Linear dielectrics and capacitors P1 School physics bridge M1 Mathematics from the beginning

1. Isolation fixes the free plate charge Q.
   Why: There is no route for charge exchange.
2. Capacitance becomes $kC_0$, so $V = Q/(kC_0) = V_0/k$.
   Why: Use the definition of capacitance.
3. $U = Q^2/(2C)$ therefore becomes $U_0/k$.
   Why: The energy formula must use the quantity held fixed.

**ANSWER** $V/V_0 = 1/k$ **and** $U/U_0 = 1/k$**; energy can be released as mechanical work.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q31 Prerequisite-course PYQ A26 Q10 · expanded

**Obtain the torque expression for a planar current loop. Then consider a 100-turn coil of area $2 \times 10^{-3}$ m² carrying 0.20 A. Its normal makes 30° with B=0.50 T. Find moment, torque and dipole energy.**

KNOW FIRST

T9 Magnetization and magnetic torque P1 School physics bridge M1 Mathematics from the beginning

1. For a rectangular loop, opposite magnetic forces of magnitude $IB\ell$ on a pair of sides form a couple. Their perpendicular separation gives $\tau = IAB\sin\theta$ for one turn, with θ the angle between its normal and B. Thus $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$, where $\mathbf{m} = IA\hat{n}$.
   Why: The Lorentz forces sum to zero in a uniform field but their moments need not cancel.

2. $m = NIA = 100(0.20)(0.002) = 0.040$ A m².
   Why: Moments of the turns add.

3. $\tau = mB\sin 30^\circ = 0.010$ N m.
   Why: Torque uses the angle of the normal, not the plane.

4. $U = -mB\cos 30^\circ = -0.0173$ J.
   Why: Choose zero dipole energy at a right angle.

**ANSWER** $m = 0.040$ **A m², $\tau = 0.010$ N m, $U = -0.0173$ J.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q32 Model · syllabus-based

**A very long cylinder has uniform $\mathbf{M} = M\hat{\mathbf{z}}$ and no free current. Find its bound currents and B far from its ends.**

KNOW FIRST

T10 Bound currents and magnetized objects P1 School physics bridge M1 Mathematics from the beginning

1. Uniform M gives $\mathbf{J}_b = 0$.
   Why: Its curl is zero.

2. On the curved surface, $\mathbf{K}_b = M\hat{\mathbf{z}} \times \hat{\mathbf{r}} = M\hat{\boldsymbol{\phi}}$; at flat ends the cross product is zero.
   Why: Use the outward surface normal.

3. The curved sheet is equivalent to a long solenoid with surface current M.
   Why: A solenoid's current loops have the same geometry.

**ANSWER** $B_{\text{in}} = \mu_0 M$ **axially and** $B_{\text{out}} \simeq 0$**, neglecting ends.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q33 Model · syllabus-based

**Inside a block, $\mathbf{M} = ax\hat{\mathbf{z}}$. Find its volume bound current and the surface current on the face x=L.**

KNOW FIRST

T10 Bound currents and magnetized objects P1 School physics bridge M1 Mathematics from the beginning

1. Only $\partial M_z/\partial x = a$ is nonzero, so $\nabla \times \mathbf{M} = -a\hat{\mathbf{y}}$.
   Why: Use the Cartesian curl component $(\nabla \times M)_y = \partial_z M_x - \partial_x M_z$.

2. At x=L, the outward normal is $\hat{\mathbf{x}}$ and $\mathbf{K}_b = aL\hat{\mathbf{z}} \times \hat{\mathbf{x}} = aL\hat{\mathbf{y}}$.
   Why: The cross-product order is M then n.

**ANSWER** $\mathbf{J}_b = -a\hat{\mathbf{y}}$; $\mathbf{K}_b(x = L) = aL\hat{\mathbf{y}}$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q34

**Prerequisite-course PYQ A26 Q16 · adapted to matter**

**A long solenoid has 1000 turns/m, current 0.50 A and a linear core with $\mu_r = 2$. Find H, M and B inside, and state the outside-field approximation. Recover the air-core case.**

KNOW FIRST

T11 The auxiliary field H P1 School physics bridge
M1 Mathematics from the beginning

1. Ampère's law for H gives $H = nI = 500$ A/m.
   Why: Only the free winding current enters.
2. $\chi_m = \mu_r - 1 = 1$, so $M = \chi_m H = 500$ A/m.
   Why: Use the specified linear response.
3. $B = \mu_0 \mu_r H = 1.257 \times 10^{-3}$ T.
   Why: B includes both free and bound-current effects.
4. For an ideal infinitely long solenoid the exterior field is zero; far from the ends of a long real one it is approximately zero. Setting $\mu_r = 1$ gives the air-core interior $B = \mu_0 nI$, while $H = nI$ and M=0.
   Why: The long-solenoid approximation neglects end fringing.

**ANSWER** $H = 500$ A/m; $M = 500$ A/m; $B = 1.257$ mT.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q35 Model · syllabus-based

**Compare P, D, M and H, and explain why $D = \epsilon_0 E$ and $B = \mu_0 H$ are not general material identities.**

KNOW FIRST

T12 Magnetic susceptibility and permeability P1 School physics bridge M1 Mathematics from the beginning

1. P is electric moment density and M is magnetic moment density.
   Why: They describe material response.
2. $D = \epsilon_0 E + P$ and $B = \mu_0(H + M)$.
   Why: These definitions retain both vacuum and material terms.
3. Dropping P or M is justified only when that response vanishes or is being neglected.
   Why: A material can contribute substantial bound charge or current.

**ANSWER In linear matter use $D = \epsilon E$ and $B = \mu H$, with specified material assumptions.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Q36 Model · syllabus-based

**Explain why a transformer core and a permanent magnet require different hysteresis behaviour.**

KNOW FIRST

T13 Ferromagnetism and memory P1 School physics bridge M1 Mathematics from the beginning

1. A transformer cycles repeatedly, so low coercivity and a narrow loop reduce hysteresis loss.
   Why: Less irreversible rearrangement occurs each cycle.

2. High permeability allows useful flux with modest excitation.
   Why: The core responds strongly to the winding current.

3. A permanent magnet needs substantial remanence and high coercivity.
   Why: It must retain its state and resist unwanted demagnetization.

**ANSWER** **Soft narrow-loop material suits transformers; hard high-coercivity material suits permanent magnets.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

## Section C

# Q37 Legacy PYQ N25 Q17 · adapted marks

**Starting from dipole potential, derive the surface and volume bound-charge representation of a polarized object and explain its physical meaning.**

KNOW FIRST

T4 Bound charge and the field of polarization P1 School physics bridge M1 Mathematics from the beginning

1. Let $\mathcal{R} = |\mathbf{r} - \mathbf{r}'|$, where r is the observation point and r′ labels source points. A volume element has dipole moment $\mathbf{P}(\mathbf{r}')d\tau'$.
   Why: Polarization is moment density.

2. Its potential sums to $V = (4\pi\epsilon_0)^{-1} \int \mathbf{P} \cdot \nabla'(1/\mathcal{R})\, d\tau'$.
   Why: The source-coordinate gradient equals $\hat{\mathcal{R}}/\mathcal{R}^2$.

3. Apply the product rule: $\mathbf{P} \cdot \nabla'(1/\mathcal{R}) = \nabla' \cdot (\mathbf{P}/\mathcal{R}) - (\nabla' \cdot \mathbf{P})/\mathcal{R}$.
   Why: This moves the derivative from the distance factor to the polarization.

4. The divergence theorem converts the first volume integral to a surface integral.
   Why: It equates outward flux and integrated divergence.

5. Thus $V = (4\pi\epsilon_0)^{-1}[\oint \sigma_b\, da'/\mathcal{R} + \int \rho_b\, d\tau'/\mathcal{R}]$.
   Why: These are exactly the potential formulas for surface and volume charge.

6. Identify $\sigma_b = P \cdot \hat{n}$ and $\rho_b = -\nabla \cdot P$. Uniform interior P cancels there, but unmatched dipole ends remain on the surface.
   Why: This explains both the signs and the physical origin.

**ANSWER** **The polarized object has the same macroscopic field as its equivalent bound surface and volume charges.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

## Q38 Model · syllabus-based

**Develop a systematic method for solving a planar dielectric interface problem with free surface charge. Include the field laws and a normal-field example.**

KNOW FIRST

T6 Electric boundary conditions P1 School physics bridge M1 Mathematics from the beginning

1. Choose normal n from medium 1 into medium 2, and resolve every field into normal and tangential components.
   Why: Sign conventions prevent ambiguous jumps.
2. A pillbox gives $D_{2n} - D_{1n} = \sigma_f$.
   Why: Its enclosed free charge is $\sigma_f A$.
3. A tiny electrostatic loop gives $E_{2t} = E_{1t}$.
   Why: Its circulation vanishes.

4. For linear isotropic media insert $D_i = \epsilon_i E_i$.
   Why: The constitutive law connects the auxiliary and physical fields.

5. If the fields are purely normal, $E_{2n} = (\sigma_f + \epsilon_1 E_{1n})/\epsilon_2$.
   Why: Solve the normal jump algebraically.

6. For $\sigma_f = 0$ and $\epsilon_2 = 3\epsilon_1$, $E_{2n} = E_{1n}/3$. Boundary conditions alone still need a source or far-field condition to fix its magnitude.
   Why: A relation between two unknown fields is not a complete solution.

**ANSWER** **Use normal D and tangential E conditions together with material laws and source/geometry information.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

## Q39 Model · syllabus-based

**Explain how microscopic dipoles lead to M, bound currents and H. Derive the free-current Ampère law and discuss its limits.**

KNOW FIRST

T11 The auxiliary field H P1 School physics bridge
M1 Mathematics from the beginning

1. Define M as the vector magnetic moment per volume.
   Why: It averages many atomic moments.

2. Microscopic loop currents cancel on internal common edges but leave a surface sheet $K_b = M \times \hat{n}$; spatial variation leaves $J_b = \nabla \times M$.
   Why: These equivalent currents reproduce the macroscopic field.

3. Start with $\nabla \times B = \mu_0(J_f + J_b)$.
   Why: All steady currents source the magnetic curl.

4. Substitute the bound current and rearrange to $\nabla \times (B/\mu_0 - M) = J_f$.
   Why: Curl is a linear operation.

5. Define $H = B/\mu_0 - M$ and apply Stokes' theorem to obtain $\oint H \cdot dl = I_f$.
   Why: Surface curl and edge circulation are equivalent.

6. For linear isotropic matter, $M = \chi_m H$ and $B = \mu H$; in a ferromagnet this single constant law generally fails.
   Why: History and saturation change the response.

**ANSWER** **H isolates free current in the magnetostatic circulation law; it does not erase the material's contribution to B.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

## Q40 Model · syllabus-based

**Compare diamagnetism, paramagnetism and ferromagnetism, including microscopic origin, field response and hysteresis.**

KNOW FIRST

T13 Ferromagnetism and memory P1 School physics bridge M1 Mathematics from the beginning

1. Diamagnetism is an induced response opposing the applied field, with small negative susceptibility.
   Why: Orbital motion changes under the field.
2. Paramagnetism aligns pre-existing moments weakly, giving small positive susceptibility.
   Why: Thermal disorder prevents complete alignment in weak fields.
3. Ferromagnetism has cooperative alignment within domains.
   Why: Interactions produce order that can persist without an applied field.
4. The initially weak net magnetization can grow by domain-wall motion and rotation.
   Why: Favourably oriented domains expand.
5. On cycling H, pinning produces hysteresis, remanence and coercivity.
   Why: The material does not follow a unique memory-free M(H) path.

6. Above the Curie temperature spontaneous order disappears; choosing a soft or hard material depends on whether repeated reversibility or retained magnetization is wanted.
   Why: Temperature and history are essential parts of the material description.

**ANSWER** **Diamagnets oppose weakly, paramagnets align weakly, and ferromagnets show cooperative, often hysteretic response.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.